

Supercomputer Report

This was my first time using CSC's supercomputer environment. The goal was to explore how a supercomputer works, run a machine learning model, and reflect on the experience.

I started by logging in through SSH and checking the Linux file system. I looked at the folders, checked where my files were stored, and practiced moving around using terminal commands.

After that, I opened a Jupyter Notebook session through CSC puhti so that I could write and run the codes in the environment. I used Python with libraries such as NumPy, pandas, and scikit-learn, loaded through CSC's module system instead of installing everything with pip.

For the machine learning model, I decided to connect the exercise to my own project, so I worked with an audio dataset related to my Silka app. I filtered the dataset to use female voice samples, extracted MFCC audio features, and trained a model to classify calm versus emotional speech. My first attempt gave me about 83% accuracy and finished in around 2 seconds; the same task took about 3 minutes on my local machine. After adjusting the feature settings and switching to a Random Forest classifier, I got the accuracy up to around 89%. My aim was not to build a perfect model, but to understand the workflow: prepare the data, train the model, check the accuracy, and find where the output files were saved.

The main difference compared with my laptop was the environment. Puhti felt much more powerful, and the notebook ran smoothly. However, it was also more technical and required the use of basic commands. On a normal laptop, I usually click folders and install packages directly. On the supercomputer, I had to use commands, load modules, and pay attention to where files were located. Small mistakes in commands could stop it from working, so I had to be more careful.

Before using Puhti, I thought supercomputers were only for advanced users only. After trying it myself, I understood that the basic workflow is possible to learn step by step and it takes practice. It was confusing at first, but I now understand the main idea better: connect to the system, load the right tools, run the code, and check the results. I feel more prepared to work on this kind of environment in future projects.